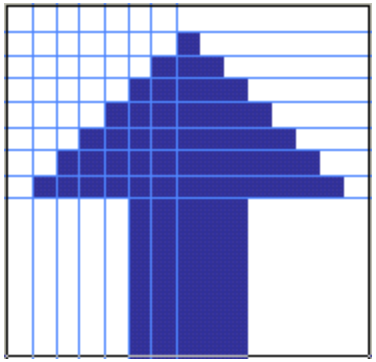
# Bitmap vs. Vector-Based Graphics

# **Bitmap** images

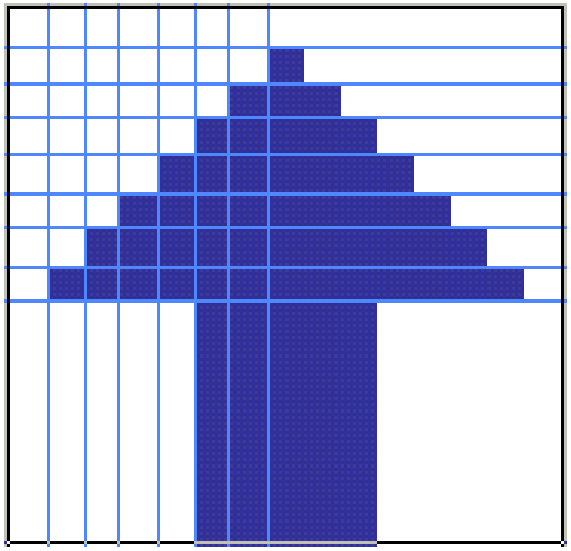
**Bitmap** images, also referred to as raster images, are pixel-based. This means that location and color information about the image is stored in individual pixels within a grid.



Arrow Image, Actual Size



Magnified Arrow Image with Pixel Grid



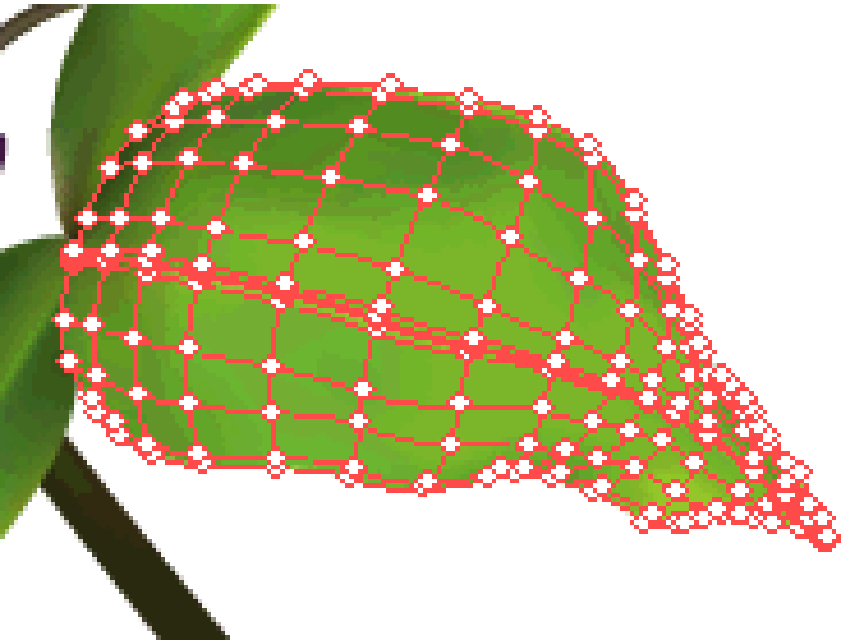
As the figure illustrates, each pixel has an assigned color; some pixels are white, while other pixels are blue. The information stored in a bitmap image regarding pixel location and color is what forms the image. Bitmap images are edited at the pixel level; in other words, the color of any one pixel can be changed. Additional attributes of bitmap images include:

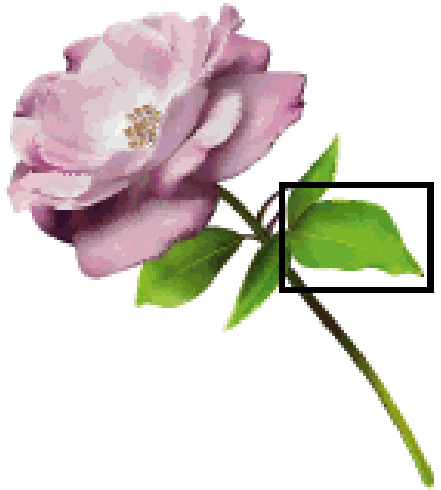
# Additional attributes of bitmap images

- Bitmap images are usually created and edited in "photo" or "paint" programs such as Adobe Photoshop.
- Bitmap images are mapped to a grid.
- The size of the image is based on the image's resolution.
- Bitmap images are not easily scalable.
- Bitmap images are used for photorealistic images and, therefore, may involve complex color variations.

# **Vector** images

**Vector** images are mathematically-based. All lines, shapes, etc. (also called objects) of a vector-based image are independent of one another





one leaf of the same rose magnified, along with the paths and handles used within vector images

# Additional attributes of vector-based images

- Vector-based images are usually created and edited in "draw" or "illustrate" programs such as Adobe Illustrator.
- Vector-based images have smooth edges.
- Vector-based images create curves or shapes.
- Vector-based images are good for precise illustrations, but are not as good for photorealistic images.
- Vector-based images are easily scalable, due to their use of mathematic formulas.

# Bit Depth

**Bit depth** refers to the number of colors that can be displayed.

The higher the bit depth, the more colors used in the image, therefore, the larger the file size.

| Bit Depth | Formula | Number of Colors |
|---|---|---|
| 1-bit | $2^1$ | 2 |
| 2-bit | $2^2$ | 4 |
| 3-bit | $2^3$ | 8 |
| 4-bit | $2^4$ | 16 |
| 5-bit | $2^5$ | 32 |
| 6-bit | $2^6$ | 64 |
| 7-bit | $2^7$ | 128 |
| 8-bit | $2^8$ | 256 |
| 24-bit | $2^{24}$ | 16,777,216 |

| Bit Depth | Number of Colors | File Size |
|---|---|---|
| 1-bit | 2 colors | 2 KB |



| Bit Depth | Number of Colors | File Size |
|---|---|---|
| 4-bit | 16 colors | 21 KB |



| Bit Depth | Number of Colors | File Size |
|---|---|---|
| 8-bit | 256 colors | 53 KB |



| Bit Depth | Number of Colors | File Size |
|---|---|---|
| 24-bit | 16.7 million colors | 224 KB |



# Resolution

**Resolution** is the number of pixels in a linear inch (i.e. pixels per inch or ppi).

More pixels means higher resolution, which creates better image quality because you end up with more realistic representations of color, better gradations of both individual colors and gray tones, and crisper images in general.

# Graphic File Formats

File formats help to identify what kind of file is being worked with and are usually distinguished by the file extension.

Most computer applications have a **native file format**

i.e., a default format for files created in that program.

Adobe InDesign saves files with an .indd extension

Adobe Illustrator saves files with an .ai extension

In addition, many applications allow the user to save files in formats other than the application's native format.

It is a good idea to save files in the native format & non-native format.